Radiative Turbulent Mixing Layers in Astrophysics

Literature Review and a Practical Framework for Computing Emission Spectra

Abstract

Radiative turbulent mixing layers (RTMLs) occur where shearing hot and cold gas mix, pass through the efficient-cooling temperature range, and radiate the enthalpy supplied by the hot phase. This note separates two related but distinct calculations: a bolometric cooling loss and an observable spectrum. It reviews the physical controls on RTMLs and presents an implementable post-processing procedure: obtain the time-dependent thermodynamic history, solve non-equilibrium ionization (NEI) where required, evaluate line and continuum emissivities from an atomic database, integrate them over volume or ray, and finally apply transfer and instrumental response. The central practical result is that a cooling table alone cannot predict a spectrum: the ion fractions and level populations must be specified. In rapidly mixed or rapidly cooled gas, the ionization/recombination time can exceed the cooling or mixing time, making CIE spectra a useful baseline but not the default prediction. For a new calculation, the most direct publicly documented implementation is the full-ionization-network extension of PLUTO; FLASH is a robust alternative but requires a new cooling/emissivity coupling, while MAIHEM is the most integrated option if its modified-FLASH source can be obtained. TORUS is better retained for transfer and synthetic-observation post-processing than used as the primary RTML microphysics solver. The locally available MAPPINGS V 5.2.1 source substantially improves the practical options: it is not a multidimensional mixing solver, but it can supply internally consistent CIE/NEI cooling histories, ion fractions, and line/continuum emissivities, as well as one-zone and radiative-shock benchmarks. The most realistic modular architecture is therefore PLUTO or FLASH for the RTML dynamics, MAPPINGS for microphysics and spectral tables, and TORUS only when a subsequent transfer calculation is required. For galactic-wind applications, the Fielding–Bryan steady multiphase-wind model provides a useful additional closure between local interface emission and the radial population of clouds, allowing a local RTML surface flux to be converted into a galaxy-scale luminosity distribution.

Contents

1. Scope and physical picture **2**

2. Development of the existing research **3**

2.1 Analytic foundations and the first spectral predictions . . . 3

2.2 The numerical transition: dimensionality, NEI, and magnetic fields . . . 3

2.3 Fractal fronts, combustion theory, and two cooling regimes . . . 3

2.4 Local front structure and observable line predictions . . . 4

2.5 High-Mach-number and magnetized extensions . . . 4

2.6 Applications and present observational status . . . 5

3. What must be calculated for a spectrum **5**

3.1 From hydrodynamics to an emissivity distribution . . . 5

3.2 Why NEI is normally necessary in RTMLs . . . 6

4. Recommended calculation procedure **6**

5. Code assessment and practical implementation **7**

5.1 Selection criteria . . . 7

5.2 Comparison of the main candidates . . . 8

5.3 Role of the local MAPPINGS V checkout . . . 10

5.4 From a local RTML to a multiphase galactic wind . . . 11

5.5 Revised practical recommendation . . . 12

5.6 A concrete production architecture . . . 13

5.7 Minimum verification matrix . . . 14

6. A minimal, useful surrogate: a steady isobaric cooling flow **14**

7. Diagnostics and observables **15**

8. Numerical and physical failure modes **15**

9. Literature context **15**

10. Conclusions **16**

1. Scope and physical picture

Consider cold gas ($T_c \sim 10^4$ K) moving relative to a hot phase ($T_h \gtrsim 10^6$ K). Kelvin–Helmholtz-driven turbulence mixes material across the interface. At nearly common pressure the mixed gas passes through roughly

$$T_{\rm mix} \sim (T_c T_h)^{1/2}, \tag{1}$$

often near 10^5 K, where metal-line cooling is efficient. In a steady radiative layer, cooling is powered principally by the enthalpy flux of entrained hot gas, with turbulent dissipation potentially contributing. The global energy statement is more robust than any particular geometric model:

$$L_{\rm rad} = \int_V n_{\rm e} n_{\rm H} \Lambda(T, x)\, dV \simeq \dot{M}_{\rm in}\, h_h + \dot{E}_{\rm diss} - \dot{E}_{\rm stored}. \tag{2}$$

Here Λ is the *net* cooling coefficient only when the assumed ionization state and radiation field match the gas. Equation (2) checks the energy budget; it does not determine how $L_{\rm rad}$ is distributed over wavelength.

The relevant competition is captured by

$$\mathrm{Da} \equiv \frac{t_{\rm turb}}{t_{\rm cool}}, \qquad t_{\rm cool} = \frac{e_{\rm th}}{n_{\rm e} n_{\rm H} \Lambda_{\rm net}}. \tag{3}$$

For $\mathrm{Da} \gtrsim 1$, cooling is fast compared with outer-scale mixing and the interface fragments into a strongly corrugated multiphase cooling front. Tan, Oh, and Gronke emphasize that the outer eddy turnover time then sets the effective consumption/mixing rate, whereas Fielding et al. find a corrugated cooling surface with an approximately $5/2$ fractal dimension. These are physical guides for convergence tests, not replacements for a spectrum calculation [8, 7].

2. Development of the existing research

2.1 Analytic foundations and the first spectral predictions

Begelman and Fabian proposed that shear-driven turbulent transport mixes hot and cold material into an intermediate phase with a characteristic temperature near $(T_c T_h)^{1/2}$, and that the incoming hot-phase enthalpy supplies the radiative loss [1]. This model established the two ideas that remain central today: the interface is an energy-conversion layer rather than an isolated parcel, and its luminosity is tied to the rate at which hot gas is entrained. Its assumed layer thickness, often written $h \sim v_{\rm turb} t_{\rm cool}$, should now be regarded as an early closure rather than a universal scaling.

Slavin, Shull, and Begelman converted this picture into emission and absorption predictions for Galactic interfaces [2]. Their calculation included time-dependent ionization and predicted C IV, Si IV, N V, O VI, UV semiforbidden lines, and Hα. This work was conceptually important because it treated ionization lag and spectra together. It also anticipated a persistent observational issue: one individual interface generally produces a modest high-ion column, so a long sightline may need to intersect many interfaces.

2.2 The numerical transition: dimensionality, NEI, and magnetic fields

Esquivel et al. performed three-dimensional MHD calculations with equilibrium cooling and demonstrated that field orientation and finite evolution time matter for the morphology and growth of a mixing layer [3]. Kwak and Shelton subsequently coupled hydrodynamics to NEI rate equations [4]. They found that mixing can transport ions into gas whose temperature differs from the CIE formation temperature, while cooling can outpace recombination. Their simulated C IV/N V/O VI ratios were broadly compatible with Galactic-halo diagnostics, but varied strongly in space and time. Thus a single static "mixing temperature" is insufficient for either ion columns or line emission.

Ji, Oh, and Masterson carried out three-dimensional HD and MHD simulations including NEI and photoionization [6]. Their results revised several early analytic expectations:

- characteristic turbulent and inflow velocities were substantially below the imposed shear velocity and only weakly dependent on it;
- the layer width followed approximately $h \propto t_{\rm cool}^{1/2}$ in their parameter regime, rather than $h \propto t_{\rm cool}$;
- ion columns retained density and metallicity dependence; and
- magnetic amplification and tension could strongly suppress mixing and reduce high-ion columns.

NEI increased the abundances and columns of high ions by factors of a few in relevant cases. Photoionization can also over-ionize the gas and modify its cooling efficiency, although Ji et al. found this secondary for the CGM conditions they explored.

2.3 Fractal fronts, combustion theory, and two cooling regimes

Fielding et al. showed that most cooling occurs in a thin, corrugated sheet whose measured area is consistent with a fractal dimension $D \simeq 5/2$ [7]. Their model connects the increase of cooling area to a hot-gas inflow scaling close to

$t_{\rm cool}^{-1/4}$ in the strong-cooling regime. The result explains why a planar interface area can severely underestimate the total emission, but the fitted fractal dimension and cutoff scale should not be treated as universal constants.

Tan, Oh, and Gronke organized the problem using turbulent-combustion theory [8]. In terms of the outer-scale turbulent velocity u' and size L, their simulations distinguish

$$Q,\ v_{\rm in} \propto u'^{1/2}(L/t_{\rm cool})^{1/2}, \qquad {\rm Da} < 1 \quad \text{(weak cooling)}, \tag{4}$$

$$Q,\ v_{\rm in} \propto u'^{3/4}(L/t_{\rm cool})^{1/4}, \qquad {\rm Da} > 1 \quad \text{(strong cooling)}, \tag{5}$$

where $Q = L_{\rm rad}/A_{\rm projected}$ is the bolometric surface brightness. The apparently conflicting $t_{\rm cool}^{-1/2}$ and $t_{\rm cool}^{-1/4}$ results in the literature can therefore describe different Da regimes. These relations predict a bolometric energy flux, not a wavelength-dependent spectrum.

2.4 Local front structure and observable line predictions

Tan and Oh made a further distinction that is especially important for the present purpose [9]. Global mass entrainment and bolometric cooling can converge when the outer turbulent scale is resolved, even if the microscopic interface is not. In contrast, the temperature PDF, ion columns, and line ratios depend on the local conductive–cooling front. A steady one-dimensional front obeys

$$\frac{d}{dx}\left(\kappa\frac{dT}{dx}\right) = j_x c_p \frac{dT}{dx} + \rho\mathcal{L}(T), \tag{6}$$

where $j_x = \rho v_x$ is the mass flux and $\rho\mathcal{L} = n^2\Lambda - n\Gamma$. The local temperature scale includes the Field length

$$\lambda_{\rm F} = \left(\frac{\kappa T}{n^2\Lambda}\right)^{1/2}. \tag{7}$$

Their 1-D front model reproduced temperature distributions, column densities, and line ratios from 3-D hydrodynamic calculations. It also showed why the temperature dependence of κ matters: Spitzer $\kappa \propto T^{5/2}$, constant conductivity, turbulent diffusion, and grid-scale numerical diffusion weight the emitting temperatures differently. This work supplies a practical subgrid model, but its published spectral tests mostly assumed CIE; an RTML spectrum should combine this local-front treatment with NEI whenever the time-scale test in Section 3.2 fails.

2.5 High-Mach-number and magnetized extensions

Yang and Ji extended plane-parallel RTML calculations to supersonic shear [10]. Below $\mathcal{M} \sim 1$, their bolometric surface brightness and intermediate-ion columns rose roughly as $\mathcal{M}^{1/2}$. Above unity they saturated, because the velocity within the cooling/mixing zone was limited by the local sound speed. Supersonic layers developed a two-zone structure: a nearly Mach-independent cooling/mixing zone and a broader turbulent zone. Turbulent dissipation, rather than hot-phase enthalpy alone, became the dominant power source and cold gas tended toward evaporation rather than condensation.

Recent MHD studies sharpen the magnetic-field caveat. Zhao and Bai found that even initially weak fields can be amplified enough to lower the surface brightness and hot-gas inflow, through both magnetic pressure and suppression of turbulent mixing [11]. Das and Gronke likewise found strong suppression in idealized plane-parallel RTMLs, while driven turbulent boxes showed smaller changes in net cold-gas growth because magnetic fields also altered global morphology and contact area [12]. Therefore an HD scaling cannot simply be multiplied by a universal magnetic correction: the Alfvén Mach number, initial topology, anisotropic conduction, and global geometry must be specified.

2.6 Applications and present observational status

In cloud–wind simulations, Gronke and Oh identified the useful survival and growth condition

$$t_{\rm cool,mix} \lesssim t_{\rm cc}, \tag{8}$$

where $t_{\rm cc}$ is the cloud-crushing time [13]. Cooling of mixed gas then creates new comoving cold material and can allow the cold mass to exceed its initial value. This is an application of RTML physics, although the curved, evolving cloud surface is not identical to a periodic plane-parallel layer.

Fielding and Bryan promoted this local cloud–wind exchange into a steady-state model for an entire multiphase galactic wind [23]. Their framework evolves the volume-filling hot phase and an embedded cold-cloud population simultaneously, treating cloud growth or destruction, drag, kinetic-energy thermalization, and metal exchange as mass, momentum, and energy source terms for the hot flow. It does not resolve an RTML, but it supplies the missing population and geometry closure needed to translate a single-interface calculation into a radial wind luminosity.

For cosmological cold streams, Mandelker et al. found that the analogous ratio of the mixed-gas cooling time to the shear-layer growth time controls whether a stream is disrupted or gains mass [14]. Their follow-up radiation model showed that KHI-induced dissipation could reach the luminosity scale of observed Lyα blobs in massive high-redshift halos [15]. That comparison remains model-dependent because Lyα requires self-shielding, resonant transfer, dust, and the external UV field; bolometric RTML power cannot be equated directly to escaping Lyα luminosity.

Taken together, existing studies support RTMLs as a physically plausible source of intermediate temperature ions and interface radiation. They do not yet establish a unique spectral template. Columns through a single interface are often below observed CGM columns, with some models requiring roughly 10^2–10^3 interfaces along a sightline [9, 6]. The number and orientation of interfaces, NEI history, magnetic topology, and radiative transfer are therefore part of the observable model rather than nuisance details.

3. What must be calculated for a spectrum

3.1 From hydrodynamics to an emissivity distribution

For an optically thin cell k, the spectral luminosity is

$$L_\lambda = \sum_k \epsilon_\lambda(T_k, n_{{\rm e}k}, f_k, Z, J_\nu) V_k, \tag{9}$$

where f is the vector of ion fractions and J_ν is any external or internally generated radiation field. It is often useful to store a differential emission measure (DEM),

$$\frac{d{\rm EM}}{d\log T} = \sum_{k\in d\log T} n_{{\rm e}k} n_{{\rm H}k} V_k, \qquad {\rm EM} = \int n_{\rm e} n_{\rm H}\, dV, \tag{10}$$

and convolve it with a CIE contribution function $G_\lambda(T)$:

$$L_\lambda^{\rm CIE} = \int G_\lambda(T; n_{\rm e}, Z)\, \frac{d{\rm EM}}{d\log T}\, d\log T. \tag{11}$$

Equation (11) is fast and excellent for a CIE reference model, but it is invalid if f is not the CIE fraction at the local temperature. Do not use a bolometric $\Lambda(T)$ as G_λ; the former has already summed over all radiative channels.
For a resolved transition $u \to l$ the volume emissivity is

$$\epsilon_{ul} = n_u A_{ul} h\nu_{ul}. \tag{12}$$

At low density, statistical equilibrium for the excited levels can usually be solved at fixed ion fraction; collisional excitation, radiative decay, recombination cascades, and (when relevant) photoexcitation enter the rate matrix. The continuum must be added on the same wavelength grid: free–free, free–bound, two-photon, and, where appropriate, dust emission. CHIANTI is well suited to optically thin UV/EUV/optical line and continuum synthesis; AtomDB/APEC is a standard choice for CIE X-ray spectra [16, 17].

3.2 Why NEI is normally necessary in RTMLs

For each element X and charge state q, evolve

$$\frac{df_{X,q}}{dt} = n_e \big[f_{X,q-1} C_{q-1}(T) + f_{X,q+1}\alpha_{q+1}(T) \tag{13}$$

$$- f_{X,q}\{C_q(T) + \alpha_q(T)\}\big] + \mathcal{P}_{X,q}(J_\nu) + \mathcal{X}_{X,q}, \tag{14}$$

with $\sum_q f_{X,q} = 1$. C, α, $\mathcal{P}$, and $\mathcal{X}$ denote collisional ionization, total recombination, photoionization, and other processes (e.g. charge exchange). Couple this stiff system to the energy equation because the emissivity and cooling rate depend on the evolving fractions. A simple decision rule is to calculate CIE only if both $t_{\rm ion}$ and $t_{\rm rec}$ are short compared with *both* $t_{\rm cool}$ and the local mixing/advection time. Otherwise use NEI.
This is not a small correction in a mixing layer: mixing can be faster than ionization, and cooling can be faster than recombination. Kwak and Shelton and Ji, Oh, and Masterson find enhanced high-ion columns in NEI relative to CIE in relevant RTML regimes [4, 6]. A spectrum intended for O VI, N V, C IV, Si IV, or soft-X-ray diagnostics should therefore report whether it is CIE, NEI, or photoionization+NEI.

4. Recommended calculation procedure

Step 1. Specify the physical experiment. Record T_c, T_h, pressure or densities, relative velocity, metallicity and abundance pattern, magnetic field/conduction assumptions, external radiation field, and the target viewing geometry. State whether the calculation is isochoric, isobaric, or fully hydrodynamic.

Step 2. Obtain a converged flow history. Run a hydrodynamic/MHD calculation with a cooling treatment consistent with the chosen abundances and radiation field, or make a one-dimensional cooling-flow surrogate. Save $(\rho, T, n_e, \boldsymbol{v})$ and either Lagrangian tracer histories or sufficiently frequent snapshots. Check that total radiative losses converge with resolution and close the energy budget in Equation (2).

Step 3. Solve the ionization history. Prefer NEI integrated along tracers (or together with the flow). Include H, He, C, N, O, Ne, Mg, Si, S, and Fe at minimum for broad UV–X-ray coverage, plus charge exchange near the cool

interface. Use an implicit/BDF solver or another positivity-preserving stiff integrator. As a controlled diagnostic, generate an otherwise identical CIE spectrum and quote the ratio line by line.

Step 4. Evaluate atomic emissivities. For each saved cell/tracer, compute level populations and ϵ_λ using its T, n_e and NEI fractions. CHIANTI/ChiantiPy is convenient for optically thin line work; AtomDB/PyAtomDB is convenient for CIE X-rays. For photoionized, optically thick, or radiation-coupled zones, use a spectral-synthesis/transfer code such as CLOUDY, but supply the non-equilibrium state or a sequence of short history steps rather than silently imposing equilibrium. The local MAPPINGS V installation is an additional, particularly useful route: precompute cooling, ionization, and emissivity tables with the same abundance and radiation assumptions used by the dynamics, or evolve representative tracer histories through its time-dependent single-zone solver.

Step 5. Project and transfer the radiation. The intrinsic surface brightness along a ray is

$$I_\lambda = \frac{1}{4\pi} \int \epsilon_\lambda(s) \exp[-\tau_\lambda(s)]\, ds. \tag{15}$$

For an optically thin forbidden line set $\tau_\lambda = 0$. Treat resonance lines (e.g. Lyα, C IV, O VI), dust UV extinction, and X-ray photoelectric absorption explicitly whenever their columns make them non-negligible. Add thermal and turbulent Doppler shifts before binning; then convolve with the telescope line-spread function and effective area.

Step 6. Validate and publish reproducibly. Compare the integrated model spectrum against the cooling loss, $\int L_\lambda d\lambda \simeq L_{\rm rad}$ after accounting for untracked bands. Test atomic-data version, time cadence, resolution, abundances, CIE versus NEI, and viewing angle. Publish line luminosities, intrinsic and attenuated spectra, and the DEM, rather than only selected line ratios.

5. Code assessment and practical implementation

5.1 Selection criteria

The relevant question is not which code has the largest collection of features, but which one minimizes the amount of new microphysics required for the intended observable. A practical RTML code should ideally provide:

1. multidimensional HD/MHD with a controllable plane-parallel shear layer;
2. passive-scalar or species advection and a stiff NEI integrator;
3. all charge states needed for C IV, N V, and O VI;
4. radiative losses evaluated from the same instantaneous ion fractions that are advected by the flow;
5. optional thermal conduction and a resolvable Field length; and
6. an export or post-processing route from cell states to line and continuum emissivities.

No candidate considered here supplies unrestricted three-dimensional dynamics, full NEI, arbitrary-geometry frequency-dependent transfer, and an observation-ready spectrum without some additional work.

5.2 Comparison of the main candidates

Code	Existing strengths	Missing or limiting feature	Recommended role
PLUTO with the Sarkar–Sternberg–Gnat extension	HD/MHD; full ion stages of H, He, C, N, O, Ne, Mg, Si, S, and Fe; ion-by-ion cooling based on CLOUDY-17; charge exchange and photoionization; public source and an NEI spectral post-processor	The published frequency-dependent transfer solver assumes spherical symmetry; the extension is based on PLUTO 4.0 and therefore needs build and regression testing before a production 3-D run	**Most direct public starting point** for a plane-parallel NEI RTML; use its 3-D ion network and optically thin spectral post-processing, not its spherical transfer solver [21]
MAIHEM	Modified FLASH with NEI, ion-by-ion cooling, photoionization and photoheating; 84 species across 13 elements; demonstrated non-CIE line synthesis with frozen ion fractions passed to CLOUDY	No clearly maintained general public distribution was identified; obtaining and validating the exact source version may require collaboration with the authors	**Best integrated physical match if the source is available** [25]
FLASH 4.8	Mature AMR HD/MHD; an `Ionize/Nei` unit that advects every charge state for a selectable set including C, N, and O; explicit/implicit isotropic and anisotropic conduction	The standard NEI unit does not feed ionization/recombination losses back into the energy equation and uses the Summers (1974) rate file; it does not produce a spectrum automatically	**Most robust fallback**: add modern ion-dependent cooling and a CHIANTI/CLOUDY emissivity layer [18, 19]
ATHENA++	Modern, efficient AMR MHD framework; already used for plane-parallel RTML calculations with cooling and conduction	Published RTML setups usually use CIE cooling and synthetic spectra rather than an in-flow full-metal NEI network	Excellent CIE dynamics baseline or a platform for a new custom NEI module, but not the shortest route to NEI emission [26, 9]

Code	Existing strengths	Missing or limiting feature	Recommended role
Standard PLUTO MINEq	NEI source integration and self-consistent optically thin cooling from 29 species; designed for emission-line diagnostics	The older network includes only the first five stages of C, N, O, Ne, and S, so it does not by itself contain O VI; superseded for this project by the full-ion extension	Useful validation case, not the preferred production network [20]
MAPPINGS V 5.2.1 (local checkout)	Thirty-element atomic setup; fixed-temperature CIE, time-dependent ionization, coupled time-dependent cooling and recombination, isochoric or isobaric evolution, detailed line/continuum emissivities, and radiative-shock models with self-consistent precursor calculations	It is a standalone, mostly interactive Fortran microphysics/spectral code, not a multidimensional turbulence or MHD solver; arbitrary 3-D RTML histories require a wrapper or precomputed tables, and one invocation per simulation cell would be impractical	**Preferred local microphysics and validation engine**: generate CIE/NEI cooling and emissivity tables, post-process representative tracer histories, and benchmark the atomic physics independently of the hydro code [22]
MULTIPHASE GALACTICWIND	Compact Python implementation of the Fielding–Bryan steady multiphase-wind model; bidirectional hot/cloud mass, momentum, energy, and metal exchange; RTML-informed cloud growth and ablation; radial cloud area and number-density closure	It neither resolves KH turbulence nor evolves NEI ions, line emissivities, MHD, conduction, or transfer; the example evolves one characteristic cloud population and uses equilibrium Wiersma et al. cooling tables	**Preferred galaxy-scale population wrapper**: convert a calibrated single-interface RTML flux into radial luminosity, surface-brightness, and velocity-profile predictions for a multiphase wind [23, 24]

Code	Existing strengths	Missing or limiting feature	Recommended role
TORUS (local checkout)	1-D/2-D/3-D AMR, Eulerian hydrodynamics, Monte Carlo radiation transport, photoionization equilibrium, dust, and line transfer; a `geometry="kelvin"` hydrodynamic initial condition is present	No ready-made collisionally ionized full C/N/O NEI network, self-consistent RTML metal cooling, MHD, or conductive front model was found; C IV is present in the local ion list, whereas N V and the higher oxygen stages needed for O VI are disabled or absent	Use as a **transfer/synthetic-observation post-processor** after importing the flow and NEI ion fractions, especially when dust, absorption, scattering, or complex viewing geometry matters [27]

5.3 Role of the local MAPPINGS V checkout

The inspected source tree is the public MAPPINGS V version 5.2.1. Its full preference set loads 30 elements, including all carbon, nitrogen, and oxygen ion stages needed for C IV, N V, and O VI. The single-zone driver provides equilibrium ionization at fixed temperature, equilibrium ionization and temperature, time-dependent ionization at either fixed or equilibrium temperature, and coupled time-dependent ionization and temperature. For the last case the density evolution may be chosen as isochoric or isobaric. The supplied cooling examples include both CIE and NEI cooling trajectories, while the cooling and spectrum routines can write element-resolved cooling, ionization fractions, line emissivities, and local or integrated spectra. These are exactly the ingredients required to construct controlled RTML spectral benchmarks. The radiative-shock module adds a second useful test. It advances the post-shock ionization and cooling zones using atomic, cooling, and flow time scales and iterates the upstream photoionized precursor [22]. Such a shock spectrum must *not* be identified with a shear-driven RTML spectrum: compression, velocity history, and self-irradiation differ. It is nevertheless a stringent test of the same ionization, cooling, and emissivity calculations over the temperature range occupied by a radiative mixing layer.

There are four recommended uses:

1. generate a version-pinned CIE cooling curve and CIE emissivity grid for the initial hydro and spectrum baseline;
2. generate isochoric and isobaric NEI cooling sequences and the associated C IV/N V/O VI contribution functions;
3. post-process a representative set of Lagrangian thermal histories from the RTML simulation; and
4. compare integrated cooling and selected line luminosities against CHIANTI, CLOUDY, and the PLUTO full-ion post-processor to expose atomic-data or normalization differences.

For CIE, a table indexed by (T, n_e, Z, J_ν) can be interpolated cell by cell. The analogous NEI table cannot generally be indexed by only the *current* (T, n_e), because two parcels at the same temperature can retain different ion fractions after different mixing and cooling histories. The production calculation must therefore do one of the following: evolve the

ions inside the hydro code; save tracer histories and pass them through MAPPINGS; or construct a deliberately restricted library of isochoric/isobaric histories and quantify the approximation. A final snapshot containing only density and temperature is insufficient for a general NEI spectrum.

Directly launching the interactive MAPPINGS executable once for every cell and output time would also be prohibitively expensive. The practical interface is a scripted batch wrapper that first computes tables on a grid of physical conditions and then uses interpolation during post-processing. If the hydro cooling is taken from those tables, the spectral calculation must retain the same abundances, dust assumptions, radiation field, and atomic-data version. A separate background metal cooling term must not be added on top of the MAPPINGS loss, because that would double count radiative channels. At the time of inspection (and again on 2026-09-27) the local source tree (`~/RT_Codes/MAPPINGS/mappings_V-521`) contained no built `map52` executable. Its documented build target is `make build`, with a default installation under `~/mappings520`. Before scientific use, build it in an isolated test location, run the supplied CIE and NEI cooling examples, and archive the exact preference, abundance, dust, and atomic-data files with the generated tables.

5.4 From a local RTML to a multiphase galactic wind

The public MULTIPHASEGALACTICWIND repository is a compact implementation of the Fielding–Bryan analytic framework [23]. It first constructs a Chevalier–Clegg-like hot wind and then solves radial ordinary differential equations for

$$\{v_{\rm w}, \rho_{\rm w}, P_{\rm w}, \rho_{\rm w} Z_{\rm w}, M_{\rm c}, v_{\rm c}, Z_{\rm c}\}.$$

The hot and cold phases exchange mass, momentum, energy, and metals. The cloud is pressure confined; its acceleration includes ram-pressure drag, momentum gained with condensing material, and gravity, while the hot wind responds to all of these source terms. This bidirectional coupling is more useful for a global wind than treating clouds as passive test particles.

The RTML closure enters through

$$T_{\rm mix} = (T_{\rm w} T_{\rm c})^{1/2}, \qquad Z_{\rm mix} = (Z_{\rm w} Z_{\rm c})^{1/2}, \qquad \xi = \frac{r_{\rm c}}{v_{\rm turb} t_{\rm cool,mix}}. \tag{16}$$

The example implementation uses a cloud-growth factor proportional to $\xi^{1/2}$ for $\xi < 1$ and $\xi^{1/4}$ for $\xi > 1$, corresponding to the weak- and strong-cooling RTML scalings, together with a separately parameterized turbulent ablation rate and an effective cooling-area boost. It therefore provides a concrete route from the local scalings in Equations (4) and (5) to a radial mass-loading model. The coefficients and density-contrast powers remain closures calibrated by idealized simulations and must be varied rather than treated as universal constants.

Its most important contribution to spectral modeling is geometrical. For a steady cloud number flux $\dot{N}_{\rm c}$ in solid angle Ω, the cloud number density is

$$n_{\rm c}(r) = \frac{\dot{N}_{\rm c}(r)}{\Omega r^2 v_{\rm c}(r)}. \tag{17}$$

If a local RTML calculation supplies a spectral surface flux $F_\lambda^{\rm RTML}$ and the effective emitting area of one cloud is $A_{\rm eff}$,

then

$$L_\lambda^{\rm c}(r) = A_{\rm eff}(r) F_\lambda^{\rm RTML}[P, T_{\rm w}, T_{\rm c}, v_{\rm rel}, Z, \text{history}], \tag{18}$$

$$\frac{dL_\lambda}{dr} = \Omega r^2 n_{\rm c} L_\lambda^{\rm c} = \frac{\dot{N}_{\rm c}}{v_{\rm c}} L_\lambda^{\rm c}. \tag{19}$$

This converts a planar or single-cloud emissivity calculation into a galaxy-scale radial luminosity. Doppler shifting each contribution by the projected $v_{\rm c}(r)$ and $v_{\rm w}(r)$ then gives an intrinsic line profile before transfer. The needed $F_\lambda^{\rm RTML}$ should come from MAPPINGS tables or resolved PLUTO/FLASH interface calculations, not from the bolometric cooling curve alone.

The repository's cooling table is based on the redshift-zero Wiersma et al. UV-background calculation [24]. It is useful for the steady wind energy equation but supplies neither the NEI state nor C IV/N V/O VI line emissivities. The code also assumes steady radial flow and a representative cloud rather than a time-dependent distribution of cloud sizes and histories. Accordingly, it should be used as a *population wrapper* around RTML microphysics, not as a replacement for the hydro, NEI, spectral, or transfer stages.

5.5 Revised practical recommendation

For work that can start from publicly identified source, the most economical choice is the Sarkar–Sternberg–Gnat PLUTO extension [21]. Its ionization network is valid over 5×10^3–3×10^8 K, includes all stages needed for C IV/N V/O VI, calculates cooling for each ion–electron pair on the fly, and supplies a post-processing tool for projected columns and spectra. The ion network is compatible with three-dimensional MHD even though its frequency-dependent radiation-transfer module is restricted to spherical symmetry. A plane-parallel RTML should therefore use the 3-D MHD+NEI part and perform optically thin spectral integration separately. The local MAPPINGS code should then be used as an independent one-zone microphysics benchmark and, where its atomic data are preferred, as a secondary emissivity-table generator.

If the MAIHEM source can be obtained and built reproducibly, it is an even closer physical match. It evolves the ionization state and computes cooling ion by ion, and published applications pass the non-equilibrium fractions to CLOUDY to calculate line emissivities. This procedure has produced strong C IV λ1549 and O VI λ1037 emission in rapidly cooling flows [25]. Its practical ranking is limited by source availability rather than physics.

FLASH 4.8 remains the safest independently maintainable alternative. Its built-in NEI unit already removes the largest bookkeeping burden—the advection and reaction solution for all C/N/O stages—but a production RTML calculation must replace or augment the old rate data and must couple the instantaneous ion state to the energy source. The documented defaults $10^4 < T < 10^7$ K and $1 < n_{\rm e} < 10^{12}\,{\rm cm}^{-3}$ also require special attention: the default lower electron-density bound is unsuitable for much of the CGM unless it is changed and validated [19].

Consequently the recommended order is:

1. **Shortest self-consistent NEI route:** the full-ion PLUTO extension for dynamics and ion evolution, with MAPPINGS and CHIANTI as independent spectral checks;
2. **most maintainable modular route:** FLASH 4.8 for AMR HD/MHD plus version-pinned MAPPINGS cooling/emissivity tables and a new coupling layer;

3. **best integrated physics if obtainable:** MAIHEM, again benchmarked against MAPPINGS;

4. **fast CIE control calculation:** ATHENA++ or standard PLUTO with a MAPPINGS/CHIANTI tabulated CIE cooling curve; and

5. **galaxy-scale population wrapper:** MULTIPHASEGALACTICWIND, after calibrating its cloud-growth, ablation, and emitting-area closures with the local RTML calculation; and

6. **radiation-transfer finishing stage:** TORUS, only after the required ion fractions and intrinsic emissivities have been supplied.

Thus the most realistic near-term combination of the available local codes is

$$\begin{aligned} \text{PLUTO/FLASH (RTML dynamics)} &\longrightarrow \text{MAPPINGS V (NEI emission)} \\ &\longrightarrow \text{MULTIPHASEGALACTICWIND (optional population model)} \\ &\longrightarrow \text{TORUS (optional transfer).} \end{aligned}$$

MULTIPHASEGALACTICWIND is omitted for a single-interface study. TORUS is omitted from the first production pass if all diagnostic lines are demonstrably optically thin and foreground attenuation can be applied analytically.

5.6 A concrete production architecture

The proposed calculation should be developed in five increasingly expensive levels.

Level 1: hydrodynamic CIE benchmark. Initialize a pressure-balanced hot/cold interface in a Cartesian box with the shear direction periodic, a transverse domain large enough to contain the growing layer, and seeded perturbations whose spectrum is recorded. Run 2-D resolution and cooling-strength scans before the 3-D production model. Compute the bolometric loss, DEM, mixing-layer width, entrainment velocity, and CIE C IV/N V/O VI columns. This stage tests Equations (2)–(5) without conflating a hydro error with an NEI error. Generate the baseline cooling and emissivity tables with MAPPINGS, and cross-check a sparse set of temperatures with CHIANTI.

Level 2: dynamically coupled NEI. Evolve at least H, He, C, N, and O as conserved species with reaction source terms. Other important coolants may either be evolved or included through a carefully separated background cooling term. The two contributions must not double count the same metal lines. Advance reactions and cooling with subcycling or an implicit stiff solver, enforce positivity and elemental normalization, and recompute n_e and the mean molecular weight from the instantaneous ion state. Compare the NEI and CIE cooling histories for identical fluid initial conditions. Validate the coupled solver against the MAPPINGS isochoric and isobaric single-zone cooling sequences before running a turbulent box.

Level 3: intrinsic emission cube and spectrum. For each transition $u \to l$, calculate

$$\epsilon_\nu(x) = n_e n_{X,q}\, q_{ul}(T, n_e)\, h\nu_{ul}\, \phi_\nu[T, \boldsymbol{v}, v_{\rm turb}], \tag{20}$$

where ϕ_ν contains thermal broadening, the cell line-of-sight velocity, and any explicitly defined unresolved velocity dispersion. Obtain q_{ul} and continuum coefficients from a version-pinned atomic data set, using either MAPPINGS or CHIANTI; sparse one-zone CLOUDY calculations with frozen NEI fractions provide a useful cross-check. If MAPPINGS is used, interpolate precomputed emissivities rather than starting one interactive model per cell. Produce both a cell emissivity cube and the volume-integrated spectrum so that inclination and aperture can be varied without rerunning the dynamics.

Level 4: optional wind-scale population synthesis. If the target is an integrated galactic wind rather than one interface, interpolate the RTML surface-flux library along the radial solution of Section 5.4. Calculate $A_{\rm eff}$, cloud number density, dL_λ/dr, projected velocity, and aperture dependence. Vary the cloud injection history or mass spectrum and the uncertain growth/ablation coefficients. Energy radiated by the interfaces must remain consistent with the mass and enthalpy exchange applied to the hot wind.

Level 5: transfer and instrument model. First calculate the optically thin result using Equation (15). Only if the inferred line-center optical depth, dust opacity, or photoelectric absorption is significant should the emission cube be passed to TORUS or another transfer solver. Importing NEI fractions into TORUS will require a defined data interface and an extension of its atomic model for N V and O VI; its current photoionization equilibrium solution should not overwrite the imported NEI state. Resonant scattering must preserve photon frequency and direction information rather than being represented by a scalar escape fraction when line profiles are an observable.

5.7 Minimum verification matrix

Before interpreting a spectrum, vary at least the grid scale, cooling normalization, conduction coefficient, magnetic-field strength/orientation, and ionization model. The minimum reported comparison is

$$\{\mathrm{CIE},\mathrm{NEI}\}\times\{\mathrm{HD},\mathrm{MHD}\}\times\{\kappa=0,\kappa=\kappa_{\rm adopted}\},$$

with matched initial conditions. Verify conservation of every elemental abundance, positivity of all fractions, cooling-energy closure, convergence of the DEM and line luminosities, and the reproduction of one-zone CIE and isochoric/isobaric NEI benchmarks. Because emissivity is approximately quadratic in density, agreement in total mass or column density alone is not a sufficient emission test.

6. A minimal, useful surrogate: a steady isobaric cooling flow

When a full 3-D RTML calculation is unavailable, a steady cooling-flow model gives a transparent first estimate. Assume a mass flux per interface area $\dot{m}/A$ and evolve a parcel from its initial mixed temperature under constant pressure. With enthalpy per mass h, solve

$$\frac{dh}{dt}=-\frac{n_{\rm e}n_{\rm H}\Lambda_{\rm net}(T,f)}{\rho},\qquad \frac{df}{dt}=R(T,n_{\rm e},J_\nu)f, \tag{21}$$

then compute a line luminosity per interface area,

$$\frac{L_{ul}}{A}=\frac{\dot{m}}{A}\int\frac{\epsilon_{ul}}{\rho}\,dt. \tag{22}$$

This has two benefits: it makes the assumed energy/mass flux explicit, and it handles ionization lag. Its limitation is equally explicit: it cannot predict the entrainment rate, corrugated area, or velocity-dependent line profile. Those require a turbulence model or simulation. Use it as a benchmark and parameter study, not as a substitute for resolved RTML dynamics. The time-dependent, isobaric option in the local MAPPINGS single-zone solver is a direct implementation path for this surrogate, while the isochoric option brackets sensitivity to the thermodynamic closure.

7. Diagnostics and observables

The expected output is multi-band. The $\sim 10^{4.5}$–$10^{5.5}$ K material can produce UV high-ion diagnostics (C IV, Si IV, N V, O VI); hotter gas contributes EUV/soft-X-ray lines and continuum; recombined material can emit optical/IR lines if it remains ionized or is externally illuminated. The classical mixing-layer models of Slavin, Shull, and Begelman explicitly connected high ions and UV/optical emission to a turbulent interface [2].
Interpret line ratios jointly with the DEM and absorption columns:

$$N_{X,q} = \int n_X f_{X,q}\, ds. \tag{23}$$

Emission scales approximately as density squared in the optically thin, low-density limit, while absorption scales linearly with density. Combining them constrains filling factor and path length far better than either alone. NEI may make an ion appear at a temperature where its CIE contribution function is small; a “formation temperature” is therefore a model-dependent description, not a direct thermometer in an RTML.

8. Numerical and physical failure modes

- **Cooling–spectrum mismatch:** a tabulated CIE cooling curve is used in the dynamics while post-processing assumes NEI or a different abundance set. Recompute or quantify the inconsistency.
- **Under-resolved numerical mixing:** numerical diffusion can set the DEM and artificially boost cooling. Demonstrate convergence of integrated luminosity, DEM, and selected line luminosities, not merely morphology.
- **Equilibrium by assumption:** CIE can miss the lagging high ions in cooling/mixing gas. Carry NEI fractions, at least in post-processing.
- **Ignoring transfer:** optically thin synthesis is not adequate for strong resonance lines or dusty/high-column sightlines.
- **Comparing unlike quantities:** distinguish intrinsic luminosity, surface brightness, attenuated flux, and detector counts; each has different geometry and units.

9. Literature context

Study	Main method	Reliable takeaway for spectral work
Begelman & Fabian (1990)	Analytic energy balance	Establishes the enthalpy–cooling balance that remains a necessary global check.
Slavin, Shull & Begelman (1993)	Turbulent-interface/NEI spectral models	Direct precedent for predicting high-ion emission and absorption from mixing layers.
Esquivel et al. (2006)	3-D MHD + CIE cooling	Demonstrates time-dependent layer growth and magnetic sensitivity; CIE limits its spectral interpretation.
Kwak & Shelton (2010)	Hydro + NEI	Mixing and cooling histories enhance high ions relative to CIE; time dependence matters.
Ji, Oh & Masterson (2019)	3-D HD/MHD + NEI + radiation	Simple $h \sim v_{\rm turb} t_{\rm cool}$ scalings can fail; NEI changes high-ion columns by factors of a few.
Fielding et al. (2020)	3-D hydro, fractal cooling front	The cooling surface is corrugated; use global energetics and resolution tests rather than planar geometry alone.
Tan, Oh & Gronke (2021)	Turbulent-combustion analogy	Da organizes weak- and strong-cooling regimes and informs convergence tests.
Tan & Oh (2021)	1-D front + 3-D spectral tests	Global entrainment and line observables have different resolution requirements; local conduction controls the temperature PDF.
Yang & Ji (2023)	High-Mach 3-D hydro	Surface brightness and ion columns saturate above $\mathcal{M} \sim 1$; dissipation can drive evaporation.
Zhao & Bai (2023); Das & Gronke (2024)	3-D MHD	Amplified weak fields suppress local mixing, while global morphology can partly compensate in turbulent systems.
Fielding & Bryan (2022)	Steady two-phase wind ODEs with RTML-informed exchange	Converts local cloud growth/ablation into bidirectional mass, momentum, energy, and metal source terms; useful for interface population and radial luminosity modeling, not for NEI microphysics.

10. Conclusions

An RTML radiation spectrum is a state-resolved emissivity calculation, not a direct plot of $\Lambda(T)$. The recommended deliverable is a NEI-based intrinsic spectrum plus transferred spectra for specified sightlines, accompanied by a DEM, ion columns, and an energy-closure test. A CIE/DEM spectrum remains valuable as a fast control model; differences from NEI then become an interpretable prediction of the mixing and cooling history.

For implementation, the current first choice is the publicly available full-ion NEI extension of PLUTO, because it already connects multidimensional MHD, the required C/N/O charge states, ion-dependent cooling, and an emission-spectrum post-processor. The local MAPPINGS V 5.2.1 checkout is the preferred complementary microphysics tool: it can supply CIE and NEI cooling/emissivity tables, one-zone cooling-flow calculations, and radiative-shock benchmarks, but it cannot form the multidimensional mixing layer. MAIHEM is the most integrated alternative if its source can be obtained, and FLASH 4.8 is the preferred robust platform when the cooling/emissivity coupling will be developed locally. The resulting practical architecture is PLUTO/FLASH for dynamics, MAPPINGS for microphysics and intrinsic radiation, MULTIPHASEGALACTICWIND when a local interface must be embedded in a radial cloud population, and TORUS only as the final transfer stage for an externally computed NEI emission cube.

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